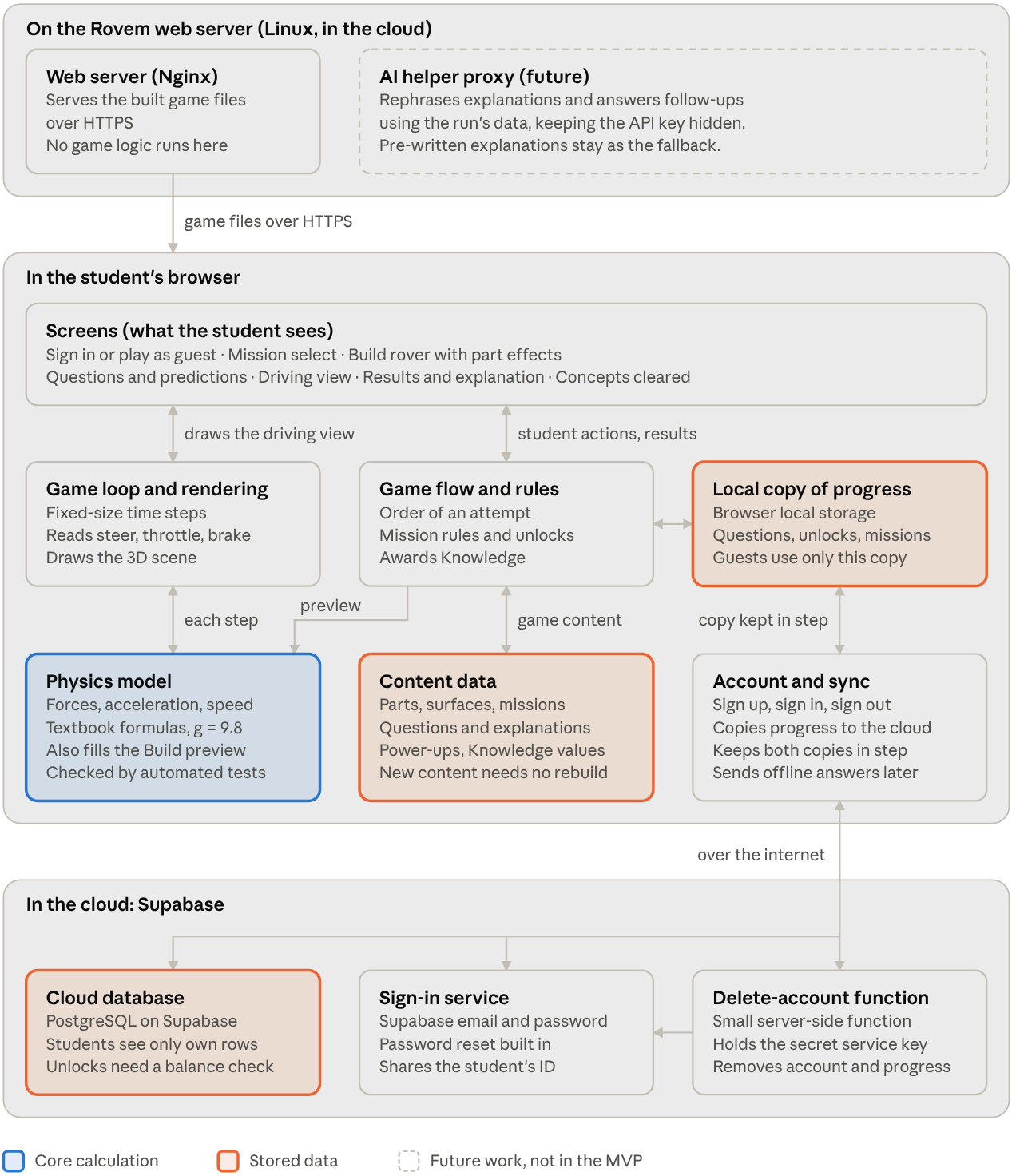
Rovem — Architecture Diagram

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Companion to the Software Requirements Specification, Draft 0.5.

Rovem runs in the desktop browser from files sent by the Rovem web server, and one physics model drives both the Build screen preview and the run. A signed-in student's progress is also saved to a cloud database; a guest's stays on the device.

Progress stays on the device; signed-in students also sync to the cloud



architecture · 11 blocks, 1 future

How to read it

Each box is one part of the game. A line between two boxes means they pass information to each other; an arrowhead shows the direction, and arrowheads at both ends mean both sides send information. The blue box is the physics. The orange boxes hold stored information: the game's content, the progress copy on the device and the cloud database. The dashed box is planned for later and is not in the first version. The three large panels show where each box runs: on the Rovem web server, on the student's own computer, or at Supabase.

Read the browser panel from the top. The student uses the screens. The screens pass the student's actions to the game flow, which decides what comes next: questions, prediction, the run, then results. To run an attempt, the game loop moves the physics model forward in small fixed time steps and draws the result. The same physics model also fills in the Build screen's preview before a run, so the numbers the student sees there match what happens when they drive.

The device always keeps a local copy of progress, and a guest uses only that copy. A student who signs in also uses Account and sync, which sends progress over the internet to the cloud database and brings it back on another device. The Supabase panel holds the database, the sign-in service and the delete-account function. That function runs in the cloud because it needs a secret key that must never reach the browser. The web server only sends the game files; later it will also run the AI helper's proxy.

What each block does

Block	What it does	SRS requirements
Web server (Nginx)	Sends the built game files to the browser over HTTPS from a Linux server in the cloud. No game logic runs on it	SRS 2.2, NFR-1, NFR-3
Screens	The pages the student sees: sign-in or guest play, mission select, build rover with part effects, question and prediction dialogs, driving view showing speed and direction, results with explanation, concepts cleared	FR-1, FR-2, FR-14 to FR-16, FR-22, FR-27, FR-32 to FR-34, FR-41

Block	What it does	SRS requirements
Game loop and rendering	Advances the simulation in fixed time steps that do not depend on frame rate, reads steer, throttle and brake, and draws the 3D scene	FR-6, FR-29, NFR-2
Game flow and rules	Sets the order of an attempt (build, required questions, predict, drive, results), applies mission rules, tracks cleared questions, awards Knowledge and applies an unlock only when the balance covers its cost	FR-5, FR-12, FR-20, FR-21, FR-26, FR-30, FR-35 to FR-40, FR-44
Physics model	Calculates driving state, net force, acceleration, speed and distance from textbook formulas with $g = 9.8\ m/s^2$, as defined in the Physics Model Specification. Also supplies the Build screen preview. Checked against hand-calculated values	FR-3, FR-8 to FR-11, FR-28, FR-29, NFR-4
Content data	Components, surfaces, terrain, missions, questions with explanations and variants, power-ups and Knowledge values, held as data so more can be added without rebuilding the game. A seeding step copies the costs, rewards and order into the database tables, and an automated test checks that the two agree	FR-19, NFR-5

Block	What it does	SRS requirements
Local copy of progress	Keeps the number of answers per question, clearing times, unlocks and completed missions in the browser's local storage, in the same shape as the database tables. Guests use only this copy. A completed mission is the safe point the game resumes from	FR-24, FR-25, NFR-7
Account and sync	Signs a student up, in and out and resets passwords. Copies a guest's progress into a new account, or asks which copy to keep if the account already has progress. Merges two devices by keeping the higher attempt count and the earlier clearing time. Sends answers and completions made offline when the connection returns	FR-25, FR-41, FR-43, NFR-7, NFR-8
Cloud database	Seven PostgreSQL tables on Supabase: three of game content, one for the player and three of progress. Each student can read and change only their own rows. One database function records an unlock only if the balance covers the cost and, for a mission, the previous mission is completed. No Knowledge total is saved; the balance is worked out from the saved answers and unlocks	FR-24, FR-25, FR-35, FR-42, FR-44
Sign-in service	Holds email-and-password accounts, sign-in and password reset. Its account ID is the student's ID in the database	FR-41

Block	What it does	SRS requirements
Delete-account function	Deletes a student's sign-in account together with all of their progress. Runs on a server because it needs Supabase's secret service key	FR-41
AI helper proxy (future)	Rephrases explanations and answers follow-up questions using the run's data. Runs on the web server behind Nginx so the API key never reaches the browser; pre-written explanations stay as the fallback	SRS Section 6